



Chapter 1

Discrete Mathematics

1.1 Objectives

Content adapted from [Piff(1991), Rosen(2007), Giunchiglia and Fumagalli(2017b)].
Please read first five chapters of [Piff(1991)] and first two chapters of [Rosen(2007)].

- ✓ 1. Defining Propositional Logic.
- ✓ 2. Defining Predicate Logic.
- ✓ 3. Defining Set, function, or relation model, and interpret the associated operations.
- ✓ 4. Operations between sets, functions, and relations.
- ✓ 5. Constructing proofs.
- ✓ 6. Logical reasoning to solve a strategic problem.
- ✓ 7. Model a real-life industry application applying formal methods.

1.2 Proposition Logic

Definition 1 (Proposition). *A proposition is a statement that is either true or false.*

1.2.1 Examples of propositions

- 1. Blue is a color.
- 2. $1 + 1 = 2$.
- 3. Any two primes have a common factor.

1.2.2 Examples of sentences that are not propositions

1. What is the color of sky?
2. $1 + 1$.
3. This sentence is false.
4. $x^2 = x$.
5. There exists an x such that, $x^2 = x$.
6. For all integers x , $x^2 = x$.

Definition 2 (Propositional variables). A proposition or a boolean variable is a variable that is either true or false.

Example 1. The following symbolic sentence is read as x squared is equal to y .

$$\underbrace{P(x, y)}_{\text{Predicates}} := \underline{x^2 = y}$$

. This is not a proposition.

However, for a given x and y , the sentence becomes a proposition.

$$P(x = 1, y = 1) := 1^2 = 1$$

1.3 Propositional calculus aka Boolean Algebra

In propositional calculus, we have different names for Boolean algebra operators.

Boolean algebra	Propositional calculus
OR gate $x + y$	Disjunction: $P \vee Q$
AND gate $x \cdot y$	Conjunction: $P \wedge Q$
NOT gate $\bar{y}$	Negation: $\neg P$

P	Q	$P \rightarrow Q$
F	F	T
F	T	T
T	F	F
T	T	T

1.3.1 Implication $\Rightarrow, \supset, \rightarrow$

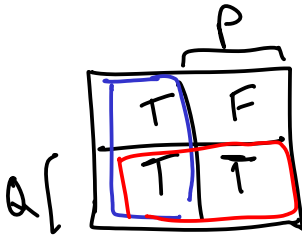
$$(P \rightarrow Q) \equiv ((\neg P) \vee Q)$$

1.4 Propositional logic language

Definition 3. Propositional alphabet

- ① Logic symbols $\neg, \wedge, \vee, \supset, \equiv$
- ② Propositional variables $p \in \mathcal{P}$
- ③ Separator symbols "(" and ")"

Definition 4. Well formed formulas



$$M_2 = \bar{P} \pm Q$$

Equivalence $\equiv \longleftrightarrow$

P	Q	$P \equiv Q$
F	F	T
F	T	F
T	F	F
T	T	T

$$(P \leftrightarrow Q) \equiv (\underbrace{\neg P \wedge \neg Q}_{m_0}) \vee (\underbrace{P \wedge Q}_{m_3})$$

- every $p \in \mathcal{P}$ is an atomic formula
- if A and B are formulas then $\neg A$, $A \wedge B$, $A \vee B$, $A \supset B$ and $A \equiv B$ are formulas.

Example 2. Let's consider a propositional language where p means "Paola is happy", q means "Paola paints a picture", and r means "Renzo is happy".

Formalize the following sentences:

1. "if Paola is happy and paints a picture then Renzo isn't happy" $(p \wedge q) \rightarrow \neg r$
2. "if Paola is happy, then she paints a picture" $p \rightarrow q$
3. "Paola is happy only if she paints a picture" $q \leftrightarrow p$

The precision of formal languages avoid the ambiguities of natural languages.

Definition 5. A Propositional interpretation is a function $I : \mathcal{P} \rightarrow \{\text{True}, \text{False}\}$.
(Evaluation of the formula with each variable set to true or false.)

Definition 6. A formula A is

Valid if for all interpretations I , $I \models A$ (All trues in TT)

Satisfiable if there is an interpretations I s.t., $I \models A$

Unsatisfiable if for no interpretations I , $I \models A$

(All F in the TT)

Propositional logic is an instrument for writing proofs. There are two connectives between two consecutive statements.

Tautology

(At least one T in the TT)

$$(p \wedge q \rightarrow \neg r)$$

p	q	r	$p \wedge q$	$\neg r$	$(p \wedge q \rightarrow \neg r)$
T	T	T	T	F	F
T	T	F	T	T	T
T	F	T	F	F	T
T	F	F	F	T	T
F	T	T	F	F	T
F	T	F	F	T	T
F	F	T	F	F	T
F	F	F	F	T	T

1.4.1 Equivalence $\iff$

Example 3 ([Giunchiglia and Fumagalli(2017a)]). Socrate says:

"If I'm guilty, I must be punished; I must not be punished." Thus I'm not guilty."

Is the argument logically correct?

$$[(p \rightarrow q) \wedge (\neg q)] \rightarrow (\neg p)$$

p	q	$p \rightarrow q$	$\neg q$	$(p \rightarrow q) \wedge \neg q$	$\neg p$
F	F	T	T	T	T
F	T	T	F	F	T
T	F	F	T	F	F
T	T	T	F	F	F

p	q	$p \rightarrow q$
F	F	T
F	T	T
T	F	F
T	T	T

p	q	$p \rightarrow q$
F	F	T
F	T	T
T	F	F
T	T	T

p	q	$\neg p$	$\neg q$
F	F	T	T
F	T	T	F
T	F	F	T
T	T	F	F

Example 4 ([Giunchiglia and Fumagalli(2017a)]). Socrate says:

"If I'm guilty, I must be punished; I must be punished. Thus I'm guilty." Is the

argument logically correct?

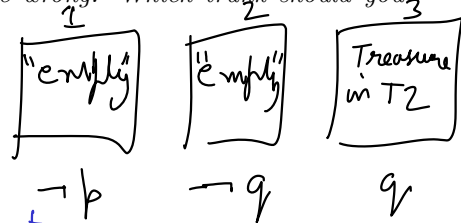
$$((p \rightarrow q) \wedge q) \rightarrow p$$

p	q	$p \rightarrow q$	$(p \rightarrow q) \wedge q$	p
F	F	T	F	F
F	T	T	T	F
T	F	F	F	T
T	T	T	T	T

not logically correct

Example 5. As a reward for saving his daughter from pirates, the King has given you the opportunity to win a treasure hidden inside one of three trunks. The two trunks that do not hold the treasure are empty. To win, you must select the correct trunk. Trunks 1 and 2 are each inscribed with the message “This trunk is empty,” and Trunk 3 is inscribed with the message “The treasure is in Trunk 2.” The Queen, who never lies, tells you that only one of these inscriptions is true, while the other two are wrong. Which trunk should you select to win?

$p = \text{“Treasure in T1”}$
 $q = \text{“Treasure in T2”}$
 $r = \text{“Treasure in T3”}$



$$\begin{aligned}
 & \underbrace{[\neg p \wedge (\neg(\neg q)) \wedge (\neg q)]}_{\text{1 correct}} \vee \underbrace{[(\neg(\neg p)) \wedge \neg q \wedge \neg q]}_{\text{2 correct}} \vee \underbrace{[\neg(\neg p) \wedge \neg(\neg q) \wedge q]}_{\text{3 correct}} \\
 & \underbrace{[\neg p \wedge (q \wedge \neg q)]}_F \vee [p \wedge (\neg q) \wedge (\neg q)] \vee [p \wedge q \wedge q] = [F] \vee [p \wedge \neg q] \vee [p \wedge q]
 \end{aligned}$$

Problem 1. Suppose that in Example 7, the inscriptions on Trunks 1, 2, and 3 are “The treasure is in Trunk 3,” “The treasure is in Trunk 1,” and “This trunk is empty.” For each of these statements, determine whether the Queen who never lies could state this, and if so, which trunk the treasure is in.

- “All the inscriptions are false.”
- “Exactly one of the inscriptions is true.”
- “Exactly two of the inscriptions are true.”
- “All three inscriptions are true.”

Problem 2 ([Giunchiglia and Fumagalli(2017a)]). Kyle, Neal, and Grant find themselves trapped in a dark and cold dungeon (HOW they arrived there is another story). After a quick search the boys find three doors, the first one red, the second one blue, and the third one green. Behind one of the doors is a path to freedom. Behind the other two doors, however, is an evil fire-breathing dragon. Opening a door to the dragon means almost certain death.

On each door there is an inscription:

freedom is be- hind this door	freedom is not behind this door	freedom is not behind the blue door
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Given the fact that at LEAST ONE of the three statements on the three doors is true and at LEAST ONE of them is false, which door would lead the boys to safety?

$p = \text{"Treasure in T1"}$

$q = \text{"Treasure in T2"}$

$r = \text{"Treasure in T3"}$

Queens

$$= \underbrace{[\neg p \wedge (\neg(\neg q)) \wedge (\neg q)]}_{\text{1 correct}} \vee \underbrace{[(\neg(\neg p)) \wedge \neg q \wedge \neg q]}_{\text{2 correct}} \vee \underbrace{[\neg(\neg p) \wedge \neg(\neg q) \wedge q]}_{\text{3 correct}}$$

$$= [\neg p \wedge \underbrace{(q \wedge \neg q)}_{\text{① correct}}] \vee [p \wedge (\neg q) \wedge (\neg q)] \vee [p \wedge q \wedge q]$$

$$= [F] \vee [p \wedge \neg q] \vee [p \wedge q] = p \wedge (\neg q \vee q)$$

$$0 + p\bar{q} + pq$$

$$= p \wedge T = p$$

Queen never lies $\Rightarrow p = T \Rightarrow \text{Treasure is in Trunk 1}$

1.5 Predicates

Definition 7 (Predicates). A predicate is a statement $P(x_1, \dots, x_n)$ involving n (not necessarily boolean) variables $x_1 \in \mathcal{X}_1$, $x_n \in \mathcal{X}_n$, and which evaluates to a proposition $P(x_1, \dots, x_n) \in \{\text{True}, \text{False}\}$.

Example 6. Let $P(x)$ denote the statement “ $x \geq 3$.” What are the truth values of $P(4)$ and $P(2)$?

1.5.1 Quantifiers

Definition 8 (Universal quantifier $\forall$).

Definition 9 (Existential quantifier $\exists$).

Example 7. Let $Q(x)$ be the statement “ $x \leq 2$.” What is the truth value of the quantification $\forall x Q(x)$, where the domain consists of all real numbers?

Example 8. Let $Q(x)$ denote the statement “ $x = x + 1$.” What is the truth value of the quantification $\exists x Q(x)$, where the domain consists of all real numbers?

1.5.2 Rules of Inference for Propositional Logic

TABLE 1 Rules of Inference.		
Rule of Inference	Tautology	Name
$\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$	$(p \wedge (p \rightarrow q)) \rightarrow q$	Modus ponens
$\begin{array}{l} \neg q \\ p \rightarrow q \\ \hline \therefore \neg p \end{array}$	$(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$	Modus tollens
$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$	Hypothetical syllogism
$\begin{array}{l} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$	$((p \vee q) \wedge \neg p) \rightarrow q$	Disjunctive syllogism
$\begin{array}{l} p \\ \hline \therefore p \vee q \end{array}$	$p \rightarrow (p \vee q)$	Addition
$\begin{array}{l} p \wedge q \\ \hline \therefore p \end{array}$	$(p \wedge q) \rightarrow p$	Simplification
$\begin{array}{l} p \\ q \\ \hline \therefore p \wedge q \end{array}$	$((p) \wedge (q)) \rightarrow (p \wedge q)$	Conjunction
$\begin{array}{l} p \vee q \\ \neg p \vee r \\ \hline \therefore q \vee r \end{array}$	$((p \vee q) \wedge (\neg p \vee r)) \rightarrow (q \vee r)$	Resolution

Example 9. Give a direct proof of the theorem “If n is an odd integer, then n^2 is odd.”

1.6 Sets

The concept of *set* is considered a primitive concept in math

A set is a collection of *elements* whose description must be unambiguous and unique: it must be possible to decide whether an element belongs to the set or not.

1.6.1 Describing sets

Listing

Abstraction

Eulero-Venn Diagrams

1.6.2 Russell's paradox

Let R be the set of all sets that are not members of themselves (sometimes called "the Russell set"). If R is not a member of itself, then its definition entails that it is a member of itself; yet, if it is a member of itself, then it is not a member of itself, since it is the set of all sets that are not members of themselves. The resulting contradiction is Russell's paradox. In symbols:

$$R = \{x \mid x \notin x\}. \text{ Then } R \in R \iff R \notin R.$$

1.6.3 Relationships between sets

Definition 10 (Empty Set $\emptyset$). $\emptyset = \{\}$

Definition 11 (Singleton $\{a\}$). $A = \{a\}$ is singleton and $\{a\} \neq a$.

Definition 12 (Set membership $a \in A$). $A = \{a, b, c\} \implies a \in A$ and $d \notin A$

Definition 13 (Set equality $A = B$). $A = \{a, b, c\}$ $B = \{c, b, a, a\} \implies A = B$.

Definition 14 (Subset $A \subseteq B$). $A = \{a, b, c\}$ $B = \{c, b, a, a\} \implies A \subseteq B$.
 $A = \{a, b, c\}$ $B = \{c, b, a, a, d\} \implies A \subseteq B$.

Definition 15 (Subset $A \subset B$). $A = \{a, b, c\}$ $B = \{c, b, a, a, d\} \implies A \subset B$.
 $A = \{a, b, c\}$ $B = \{c, b, a, a\} \implies A \not\subset B$.

Definition 16 (Power set $P(A)$, 2^A). $A = \{a, b, c\}$, then $P(A) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\}\}$

Definition 17 (Union of two sets $A \cup B$). $A = \{a, b, c\}$ $B = \{c, b, a, a, d\}$
 $\implies A \cup B = \{a, b, c, d\}$. $A \cup B = \{e \mid (e \in A) \vee (e \in B)\}$

Definition 18 (Intersection of two sets $A \cap B$). $A = \{a, b, c\}$ $B = \{c, b, a, a, d\}$
 $\implies A \cap B = \{a, b, c\}$. $A \cap B = \{e \mid (e \in A) \wedge (e \in B)\}$



elements

$$A = \{a, b, c\}$$

Power set = set of all possible subsets of A

$$P(A) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\}\}$$

$$n = |A| = \text{Cardinality of the set } A$$

$$|P(A)| = 2^n$$



1. Order or repetition of elements in the set does not matter.

1.7. RELATIONS

Set minus

$$B - A = \{d\}$$

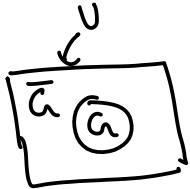
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Definition 19 (Set difference $A \setminus B$, $A - B$). $A = \{a, b, c\}$ $B = \{c, b, a, a, d\}$

$$\Rightarrow A \setminus B = A - B = \{a\} = \emptyset.$$

$$B \setminus A = B - A = \{d\}.$$

$$A \setminus B = A - B = \{e \mid e \in A \wedge e \notin B\}$$



$$\bar{A} = U - A$$

Definition 20 (Set complement $\bar{A}$, $C_U A$). Given a universal set U , complement is defined as $\bar{A} = C_U A = U - A$. Also, $A \cup \bar{A} = U$.

Example 10. Prove that $U = (A \cap B) \cup (\bar{A} \cap \bar{B})$

Example 11. $A = \{a, e, i, o, \{u\}\}$ and $B = \{i, o, u\}$ $\{u\} \neq u$

False 1. $B \in A$ B is an element of A / B belongs to A
 2. $(B - \{i, o\}) \in A$ True B is a member of A

$\{a, i\}$ 3. $\{a\} \cup \{i\} \subseteq A$ True

True 4. $\{u\} \in A$ Proper subset

5. $\{\{u\}\} \in A$ False

$\{u\}$ 6. $B - A = \emptyset$ False

7. $i \in A \cap B = \{i, o\}$ True

8. $\{i, o\} = A \cap B$ True

$$A = \{a, e, i, o, \{i, o, u\}, \{u\}\}$$

$$B \in A$$

$$B - \{i, o\} = \{u\} \notin A$$

$$\{\{u\}\} \subseteq A \text{ True}$$

subset

Example 12. Prove De Morgan laws for sets $\overline{A \cap B} = \bar{A} \cup \bar{B}$, and $\overline{A \cup B} = \bar{A} \cap \bar{B}$.

$$e \in A \cap B = e \in (A \cap B) = (e \in A) \wedge (e \in B) \quad (1.1)$$

$$e \in \overline{A \cap B} = \neg((e \in A) \wedge (e \in B)) = (\neg(e \in A) \vee \neg(e \in B)) = (e \notin A) \vee (e \in B) \quad (1.2)$$

$$e \in (\bar{A} \cup \bar{B}) = (e \in \bar{A}) \vee (e \in \bar{B}) = (e \notin A) \vee (e \notin B) \quad (1.3)$$

1.7 Relations > Functions

Definition 21 (Cartesian product of sets $A \times B$). $A \times B = \{(a, b) : \forall a \in A, \forall b \in B\}$

tuple

Definition 22 (Relation $R \subseteq A \times B$). We write aRb to read as "a is R-related" for $(a, b) \in R$ to b".

5

$$A = \{a, e, i, o, \{u\}\}$$

$$B = \{i, o, u\}$$

3

$$A \times B = \{ \underbrace{(a, i)}_{\text{tuple of a, i}}, (a, o), (a, u), (e, i), (e, o), (e, u), \dots, (\{u\}, i), (\{u\}, o), \dots \}$$

order matters

$$|A \times B| = |A| |B|$$

$$A \times B \neq B \times A \quad B \times A = \{(i, a), (o, a), (u, a), \dots\}$$

Boolean algebra $\Leftrightarrow$ Proposition calculus

Set algebra $\swarrow$

$$A = \{x \mid P_A(x)\}$$

$\downarrow$
 $x \in U$

$$B = \{x \mid P_B(x)\}$$

$\downarrow$
 $x \in U$

$$\bar{A} = \{x \mid \neg P_A(x)\}$$

$$A \cap B = \{x \mid P_A(x) \wedge P_B(x)\}$$

$$A \cup B = \{x \mid P_A(x) \vee P_B(x)\}$$

De Morgan's theorem of set algebra

$$\overline{A \cap B} = \{x \mid \neg(P_A(x) \wedge P_B(x))\}$$

$$= \{x \mid (\neg P_A(x)) \vee (\neg P_B(x))\}$$

$$= \{x \mid \neg P_A(x)\} \cup \{x \mid \neg P_B(x)\}$$

$$= \bar{A} \cup \bar{B}$$

$$\overline{A \cup B} = \bar{A} \cap \bar{B}$$

Example 13

$$\text{Dom}(R) = \{3, 5\}$$

$$\text{Im}(R) = \{6, 12, 10\}$$

$$R^{-1} = \{(6, 3), (12, 3), (10, 5)\}$$

Example 13. given $A = \{3, 5, 7\}$, $B = \{2, 4, 6, 8, 10, 12\}$ and aRb iff a is a divisor of b , then $R = \{(\underline{3}, \underline{6}), (\underline{3}, \underline{12}), (5, 10)\}$

Definition 23 (Domain of R). $R \subseteq A \times B$

Definition 24 (Image of R).

Definition 25 (Inverse of R).

$$\text{Dom}(R) = \{a \mid (a, b) \in R\}$$

$$\text{Im}(R) = \{b \mid (a, b) \in R\}$$

$$R^{-1} = \{(b, a) \mid (a, b) \in R\}$$

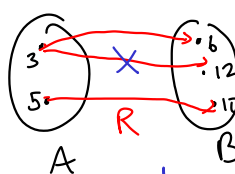
1.8 Functions $\subset$ Relation

(mapping)

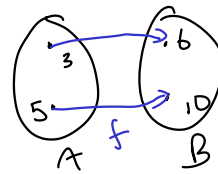
Definition 26 (Function $f: A \rightarrow B$). Given two sets A and B , a function f from A to B is a relation that associates to each element a in A exactly one element b in B . Denoted with $f: A \rightarrow B$

$$f = \{(3, 6), (5, 10)\}$$

valid function



Invalid function



1.8.1 Classes of Functions

Definition 27 (Surjective).

Definition 28 (Injective).

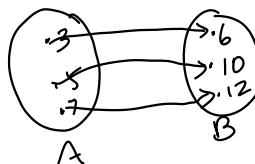
Definition 29 (Bijective).

Definition 30 (Inverse).

Definition 31 (Composition).

Surjective function

Every element in B must be mapped from some element in A



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